

Stemy — Milestone 2 Developer Handoff

Project: Stemy AI Mastering Platform **Site:** <https://stemy-frontend.vercel.app/pages/home.html> **Milestone:** Milestone 2 (held in Fiverr escrow until items below are complete) **Date:** May 2026

How to use this document

This is the complete list of everything needed to close out Milestone 2 and move into Milestone 3 (launch prep). Items are tagged:

- **P0** = Blocking M2 approval — must be done before I release the \$350 escrow
- **P1** = Functionality status check — confirm each works end-to-end
- **P2** = Cleanup polish — should be done before public launch
- **NEW** = Brand new asset or addition to the site

I've attached the screenshots and reference files. Please work through this list and let me know when items are complete so I can verify and approve M2.

1. P0 — BLOCKING Milestone 2 approval

These must be done before I approve M2 and release the \$350 escrow payment.



CRITICAL — Master flow fails / terminates mid-process

This is the **most important issue on this entire list**. The core feature of the product isn't working.

- **Master flow crashes mid-process with multiple failure modes:**
 - **Variant 1:** Click "Master Track" → progress bar starts → crashes BEFORE reaching 22% → page redirects back to Studio with no download, no error message
 - **Variant 2:** Processing appears to complete → starts downloading → download stops mid-way with "**terminated**" message → no file delivered
 - Both reproduce consistently on uploaded files

- The fact that it crashes so early (under 22%) suggests the failure is in the analysis/initialization phase, NOT in the heavy DSP processing — likely candidates: file read step, LUFFS analysis, format detection, or memory allocation
- **Required fixes:**
 1. Find and fix what's crashing the backend processor early in the pipeline (likely the analysis or file-read step, not the limiter)
 2. Investigate server timeouts for the "terminated" variant — connection is dying before the file finishes sending. Check max execution time, response size limits, and streaming setup
 3. Add proper error handling — if processing or download fails, show the user a clear error message (e.g. "Processing failed — please try again or contact support") instead of silently redirecting or terminating
 4. Add server-side logging at every pipeline step so future failures can be debugged — log: upload received, file validated, analysis started, processing started, encoding finished, download served
 5. Add a "Retry" button when processing fails so the user can try again without re-uploading
 6. Consider using a job queue + signed download URLs so large file generation doesn't block the request and downloads come from a stable URL that won't terminate
- **This is the core feature of the entire product — must work 100% before M2 approval**
- **Note for dev:** Please test the master flow with each file format separately (MP3, WAV, AIFF, FLAC) and report which formats reproduce the crash. If it crashes on all formats, the issue is before the file is even read. If it crashes only on certain formats, the issue is in the decoder/format handler.

Home page copy fixes:

-  **FALSE MARKETING CLAIM in meta description** — currently reads: *"Stemy - Professional AI-powered audio mastering built by Grammy-winning engineers."* Choo is **platinum-certified**, NOT Grammy-winning. This is a false claim that could expose me to liability and damage credibility if discovered.
- **Fix:** Change to *"platinum-certified producer"* or *"platinum engineers"* to match the rest of the site (which correctly says "platinum producer" / "platinum-certified" / "30+ years")

- Check every page on the site for "Grammy" mentions and remove or correct them all

Email consistency:

-  **Contact email mismatch on home page FAQ section**
 - FAQ "Email us" link currently goes to `hello@stemy.app` ❌ (wrong domain AND wrong username)
 - Should be `info@getstemy.com` (matches the footer Contact Us link and the official email)
 - Audit every email reference site-wide and replace any `stemy.app` or `hello@` addresses with `info@getstemy.com`

Home page "Try It Yourself" preview fixes:

-  **No audio plays on the 30-second preview sample** — even when a user successfully uploads and the preview generates, no audio actually plays back. The playback functionality is broken.
 - Test: upload a working MP3 → wait for preview to generate → click Play → nothing happens (or possibly silent player)
 - Could be: audio source not wired to player, missing audio file, broken Web Audio API setup, CORS issue, missing autoplay handling, or the preview generation step never produces an audio output file
 - **Required fixes:**
 1. Confirm the preview generation step actually produces an audio file the player can load
 2. Wire that audio file URL into the HTML5 `<audio>` element or Web Audio source
 3. Add error handling: if audio fails to load, show user "Preview failed to load — please try again" instead of silent failure
 4. Add server-side logging for preview generation success/failure
 5. Test on multiple browsers (Chrome, Safari, Firefox) — audio autoplay rules differ by browser
- **Preview upload fails on full WAV files** — current 20 MB cap blocks most WAV uploads since users naturally drop their full song, not a pre-trimmed clip
 - **Fix: trim audio client-side to first 30 seconds before upload**

- Browser uses Web Audio API to read the file, extract the first 30 seconds, and upload only that small clip
- User drops any-size file (WAV/AIFF/FLAC/MP3), system handles the 30-second extraction automatically, no error shown
- Goal: user drops their full song → preview plays back mastered version → no walls, no bounce
- This is how LANDR and eMastered handle previews — standard pattern
- File format pills shown below the upload button should match what's actually accepted:
WAV · AIFF · FLAC · MP3

Genre preset selection fixes:

-  **Trap preset button always shows the orange glow as if it's pre-activated — DONE, verified**
 - Originally looked like Trap was selected by default even when the user hadn't picked it
 - Now: orange glow + fire emoji only appear when the user has actively selected Trap
 - This same selection behavior should work for all 8 presets (Pop, Hip-Hop, R&B, Rock, Electronic, Acoustic, Country, Trap) — only the active one shows the highlight (still needs verification across all 8)

Login page fixes:

- **Dead links** — "Forgot password?", "← Back to log in", and "← Change email" all currently point to # and don't trigger the correct modal/screen. Wire these up to the proper flow steps.
- **"Studio" nav link** at the top has no URL attached — needs to link to the Studio page

Site-wide metadata cleanup:

- **All meta tags reference** `stemy.com` but the official domain is `www.getstemy.com` — replace every instance:
 - `og:url` should be `https://www.getstemy.com` (or the current Vercel preview URL during dev)
 - `og:image` should resolve to a real image hosted on `getstemy.com` (not `stemy.com`)
 - Twitter card meta tags should match

- This breaks social media previews when sharing — fix sitewide, not just login page
- **Footer brand line:** *"built by a platinum music producer with 30+ years in the game"* — confirmed accurate, **keep as-is** (30+ years is correct)
- **Contact email:** Confirm `info@getstemy.com` is wired up everywhere a Contact Us link/button appears

Studio page fixes:

-  **VU meter centering — SITE-WIDE FIX NEEDED** — VU meter graphic sits too far to the left in its container across multiple pages:
 - Studio page VU meter
 - Home page 30-second preview VU meter
 - **New SSL-9000 console VU meters** (each channel — also off-center after integration)
 - Any other VU meter on the site
 - **The fix needs to be applied uniformly across all pages** — please don't fix just one and leave the others. Use a shared CSS class so they all behave the same.
 - **Fix:**
 - Wrap each VU meter SVG in a flex container with `justify-content: center; align-items: center`
 - SVG itself should have `display: block; margin: 0 auto`
 - Check that parent containers aren't using `align-items: stretch` or other flex behaviors that override centering
 - The asymmetric arc (more space on negative side than positive) creates an optical illusion of being off-center — compensate by shifting the SVG pivot point slightly to the right, OR use a more symmetric arc
- **VU meter curved numbers** must match home page (-20 -10 -7 -5 -3 -2 -1 0 +1 +2 +3 along the arc)
- **New console VU meter — fix 0 / +3 label spacing on the right side** — the "0" and "+3" labels are too close together / overlapping. Spread them apart so each is clearly readable
- **"Decoding your track..." loading text** — hide by default, only show during actual processing

- **Master Complete results** must show real LUFS/dBTP values from actual analysis (currently hardcoded to -14 LUFS · -1 dBTP · 9.2 DR · 88s)
- **File format consistency** — display as “WAV · AIFF · FLAC · MP3” (remove duplicate AIFF, remove M4A)
- **Remove “Smart Auto-Tune Active”** badge

[INSERT SCREENSHOT: Current Master Complete page showing hardcoded values]

[INSERT SCREENSHOT: Studio page VU meter — current vs. correct]

Brand fixes (apply EVERYWHERE on the site):

- Change all instances of “Choo The Specialist” to “**Choo The Specializt**” (with a Z, ending “izt”)
- This is his actual professional name, not a typo — he’s been using it for 15+ years

2. P1 — Functionality status check

I need confirmation each of these works end-to-end. I’ll be testing on my computer once you confirm:

Account & Auth:

- Google signup creates real account
- Email signup creates real account with welcome email
- Terms of service checkbox required on signup
- Password confirm field on signup
- Forgot password sends real reset email
- Profile editing persists (name, email, password change)

Stripe & Subscriptions:

- **Stripe sandbox VERIFIED working** (May 14, 2026) — full payment loop tested with test card 4242 4242 4242 4242
 - Checkout → 7-day trial → confirmation page → account flips to TRIALING → \$9.99 Basic plan shown → May 21 2026 next billing → Invoice History pulls from Stripe (\$0.00 trial entry)

- Basic (\$9.99/mo) plan checkout works in production
- Pro (\$24.99/mo) plan checkout works
- Cancel subscription flow works
- Upgrade Basic → Pro works
- Downgrade Pro → Basic works
- Invoice History pulls real data from Stripe in production

[INSERT SCREENSHOTS: Stripe payment flow — checkout, confirmation, dashboard showing TRIALING]

Master Flow:

- Upload audio file → process → download flow works end-to-end
- Output WAV/AIFF/FLAC/MP3 file plays correctly
- **User-supplied metadata gets embedded into the downloaded file:**
 - Album cover art image (the user uploads this on the studio page) — must appear in the file's metadata so it shows up in iTunes, Spotify uploads, etc.
 - Title field
 - Artist field
 - Album field
 - Year field
 - Genre (based on selected preset)
 - "Mastered by Stemy" / "Mastered by Choo The Specializt" should also be included
- Auto-detected metadata also embedded (BPM, key if detected)
- Verification: download the mastered file → open in iTunes / Apple Music / a tag editor → confirm cover art and all fields show correctly
- Notification emails fire correctly (master complete, processing started, etc.)
- All 8 genre presets actually pass different parameters to the audio engine (not just visual selection)
- Trap preset is the flagship — should be visually highlighted with fire emoji and orange glow

3. P2 — Cleanup polish

Lower priority but should be done before public launch:

- **Genre preset order inconsistency** — list appears in different orders in different places:
 - **Style picker on home page Try It Yourself:** Pop, Hip-Hop, R&B, Rock, Electronic, Acoustic, Country,  Trap
 - **Pricing section:** Pop, Hip-Hop, R&B, Rock, Electronic, Country, Acoustic, Trap
 - Pick ONE order and apply everywhere (recommend matching the style picker order)
- **Verify these linked pages actually exist** — confirm each loads, or build them if missing:
 - `/pages/about.html` (linked from “About Choo” footer link AND the hero badge)
 - `/pages/privacy.html` (Privacy Policy)
 - `/pages/terms.html` (Terms of Service)
 - `/pages/mastering.html` (Studio link)
- **Footer “Online Mastering” and “Stem Mixer”** — both currently point to the same URL (`/pages/mastering.html`). They should go to different destinations:
 - “Online Mastering” → main mastering tool / Studio page
 - “Stem Mixer” → Stem Mixing & Mastering tab specifically (or the Pro plan view of the Studio page where the new console lives)
- Fix dead footer links — Blog currently goes to `#`, Mastering API goes to `#pricing`
- **Footer “About Choo” link** — confirm the linked page exists, build it if it doesn't
- All 8 genre presets visible in pricing section: Pop, Hip-Hop, R&B, Rock, Electronic, Acoustic, Country, **Trap**
- Pro features showing “Coming Fall 2026” badges where appropriate
- Auto-Tune and Reverb-Delay show “Coming Soon” pills
- “Mix & Master Your Sound” subheading visible on homepage
- A/B player works on Studio page
- Metadata section collapsed by default on Studio page

- Footer cleaned up (no broken links, no dev placeholders)
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4. NEW — Add Choo's plaques to credibility section

I'm sending you photos of Choo's gold record and 4 platinum plaques. These need to go on the site to back up his credentials.

Where to display:

- Add a "Credentials" or "Track Record" section on the About/Choo bio area
- Display all 5 plaque photos in a clean grid or carousel
- Each plaque should have a caption naming the artist/album (I'll send the captions with the photos)

What's coming:

- 1 gold record plaque photo
- 4 platinum plaque photos (Tupac, Ghostface Killah/Wu-Tang Clan, Trillville, [4th artist])
- Captions for each

[INSERT PHOTOS: Gold + 4 Platinum plaques — coming separately]

Image specs:

- Optimize the photos for web (compress, don't upload raw camera files)
 - Lazy load if they're below the fold
 - Make them clickable/expandable so visitors can see the details
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5. DSP Engineer Handoff Prep — Set up the audio engine for handoff to a DSP specialist

After launch, I'm bringing on a dedicated DSP audio engineer to tune the genre presets using reference WAVs from Choo. To make that handoff clean, please structure the audio code as follows:

- **Modular preset files** — one file per genre preset (e.g. `presets/trap.py`, `presets/hiphop.py`, `presets/pop.py`, etc.)

- **Configurable parameters** in each preset (EQ frequencies, comp ratios, limiter ceiling, saturation amount, etc.) so the DSP engineer can tune values without touching core code
- **Documented DSP chain** — the processing order clearly noted in code comments:
 - Input EQ → Tube Saturation → Bus Compression → Stereo Imager → Brickwall Limiter
- **Real audio analysis output** — LUFS, dBTP, dynamic range, length (no more hardcoded values)
- **Test hooks** — way to A/B test a preset before/after a code change
- **README** in the audio processing folder explaining how to add new presets, the parameter structure, and how to run tests
- **Stable frontend ↔ audio API** so the DSP engineer can change the audio side without breaking the frontend

Important: Please do NOT request the reference WAVs from me. Those are going to the DSP engineer directly when I bring them on. Just set up the architecture so they can plug in cleanly.

6. Open questions — please answer

1. **What is the latest deploy URL?** My browser keeps caching the old version
 2. **Can you disable Vercel caching during dev** so I can see your updates in real time?
 3. **Account ownership** — at the end of Milestone 3, I need everything transferred to my accounts:
 - Vercel deployment
 - GitHub repo
 - Stripe (I'll create my own and provide live API keys)
 - Any other third-party services
 4. **What stack are you using on the backend?** (For my future DSP engineer's reference)
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7. Milestone 3 Preview (for your planning)

Once Milestone 2 is approved, Milestone 3 will cover:

- Custom domain setup
 - Switch Stripe from sandbox to live mode (I'll provide live keys after I connect my bank — currently in the 1-2 day verification)
 - Real assets in place (Choo's plaque photos, final logo files, etc.)
 - Account ownership transfer (Vercel, GitHub, etc.)
 - DSP architecture handoff documentation
 - Final QA pass
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Communication

Please use Fiverr messages for updates. If you need to share long technical docs or code, Google Docs works better than Fiverr's message limit.

Looking forward to wrapping this up clean. Let me know when items are ready for verification.

— Vance